

BAG & BREAK: The Architecture of Orbital Extraction

*A Defensive Blueprint and
Technical Manifesto for the
Post-Asteroid Paradigm.*



The Post-Asteroid Paradigm:

Earth transitions from an industrial foundry to a "Luxury Resort Planet."

The Math of Disruption:

A single 200m asteroid yields 10–20M tons of material.

The Flow Rate:

1,000 tons/yr Platinum Group Metal (PGM) delivery = 5x current global supply.

Machin fabrics and charcoal carbonaceous regoliis structures.

The Inevitability:

Total commodity market repricing.
Global supply chain recalibration.



The Microgravity Mining Paradox:

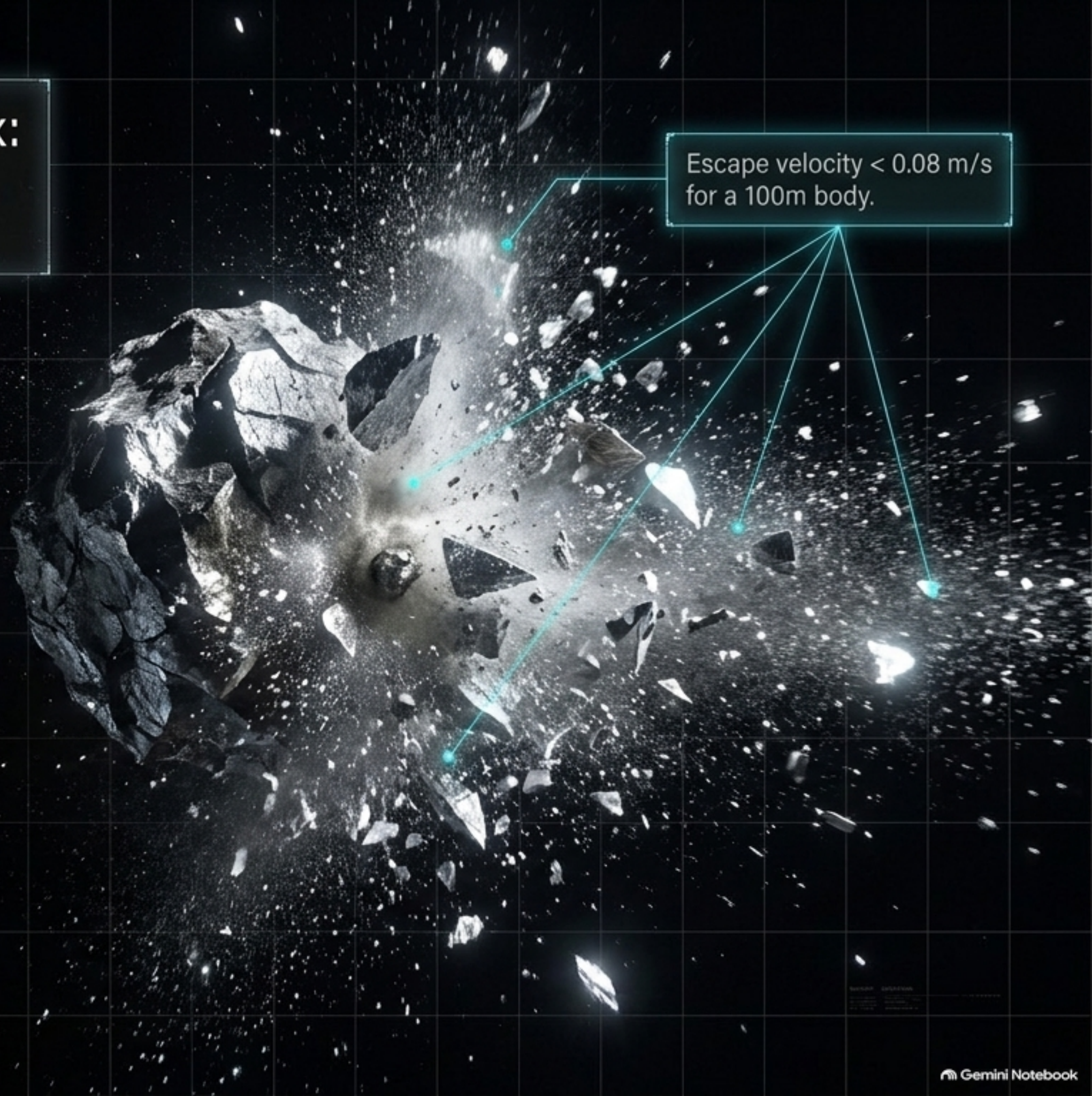
The core problem isn't breaking the rock;
it's fractionation in zero gravity.

The Hazard:

Smash it, and you create a lethal,
uncatchable debris swarm.

The Axiom:

"Crack it like a geode, not a riot."
Containment precedes disaggregation.



Escape velocity < 0.08 m/s
for a 100m body.

Target Archetypes: The Anatomy of the Prize

C-Type
(Carbonaceous /
Rubble Pile)



Density

~1300 kg/m³

Cohesion

25 Pa
(Weak)

Escape Vel

0.043 m/s

Fracture
Predictability

Low
(Chaotic, pre-broken)

M-Type
(Metallic Core)



~5400 kg/m³

300 Pa
(Solid)

0.087 m/s

High
(Directional, clean)

Conclusion: M-Type is the 'Jackpot.' High energy to fracture, but predictable, controllable mass fractionation.

The Diagnostic Void: The Problem of the Regolith

The Hurdle: We cannot mine what we cannot characterize. Surface dust hides the structural truth of the core. How do we ask a rock what's inside without a \$300M drill rig?

The Axiom: You don't ask the asteroid for permission. You stab it in the heart and listen.

SONODART™ (The Kinetic Stethoscope)

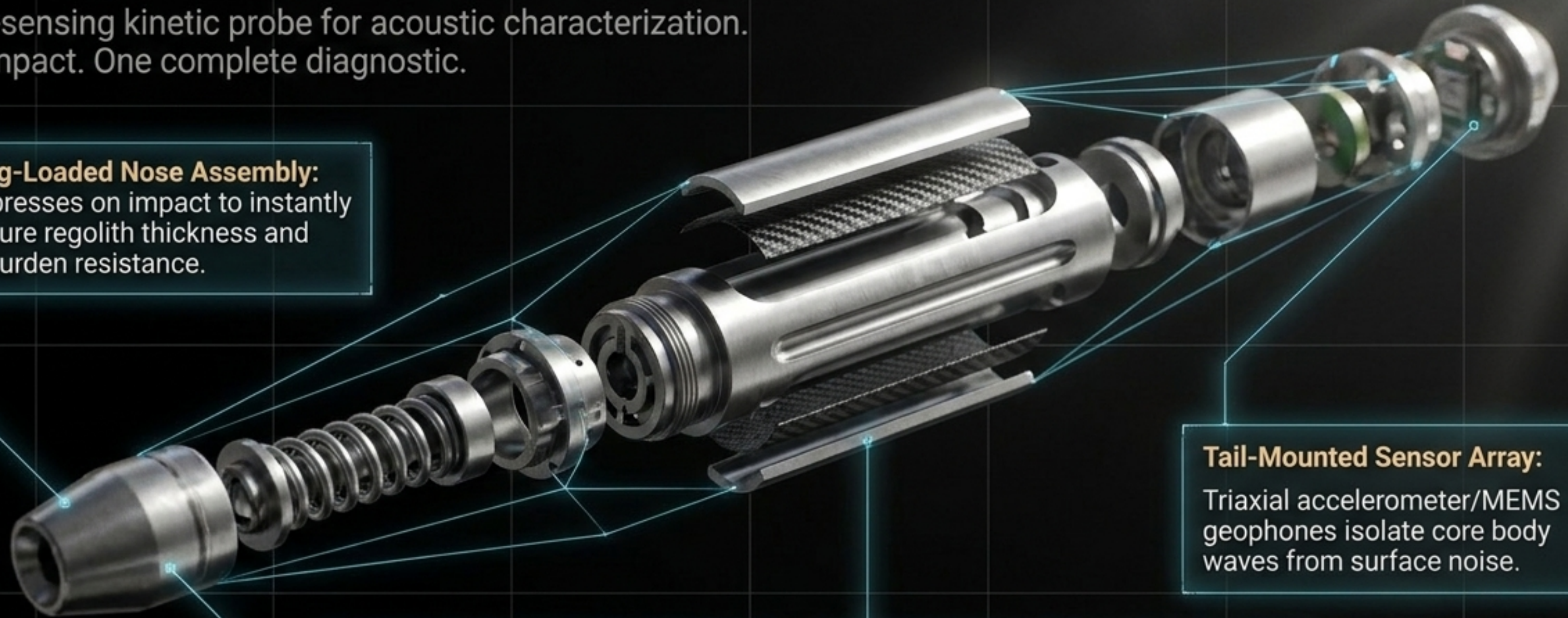
A self-sensing kinetic probe for acoustic characterization.
One impact. One complete diagnostic.

Spring-Loaded Nose Assembly:
Compresses on impact to instantly measure regolith thickness and overburden resistance.

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Tungsten Waveguide Shaft:
Tuned monolithic body to mechanically amplify and filter specific acoustic resonance frequencies.

Tail-Mounted Sensor Array:
Triaxial accelerometer/MEMS geophones isolate core body waves from surface noise.



The 'Ting' Test Framework: "If it Tings, it Sings."

Step 1: Impact
-> Kinetic impulse delivered.

Step 2:
Acoustic Signature ->
Waveform Analysis.

Branch A: THUD

High attenuation, chaotic reflections. Structure is a rubble pile.

Action: Abort or switch to chemical digestion.

Branch B: TING

High cohesion, low damping, distinct standing waves. Structure is crystalline/M-Type.

Action: Proceed to Bag & Break.

The Containment Imperative (Bagging)

The Rule:

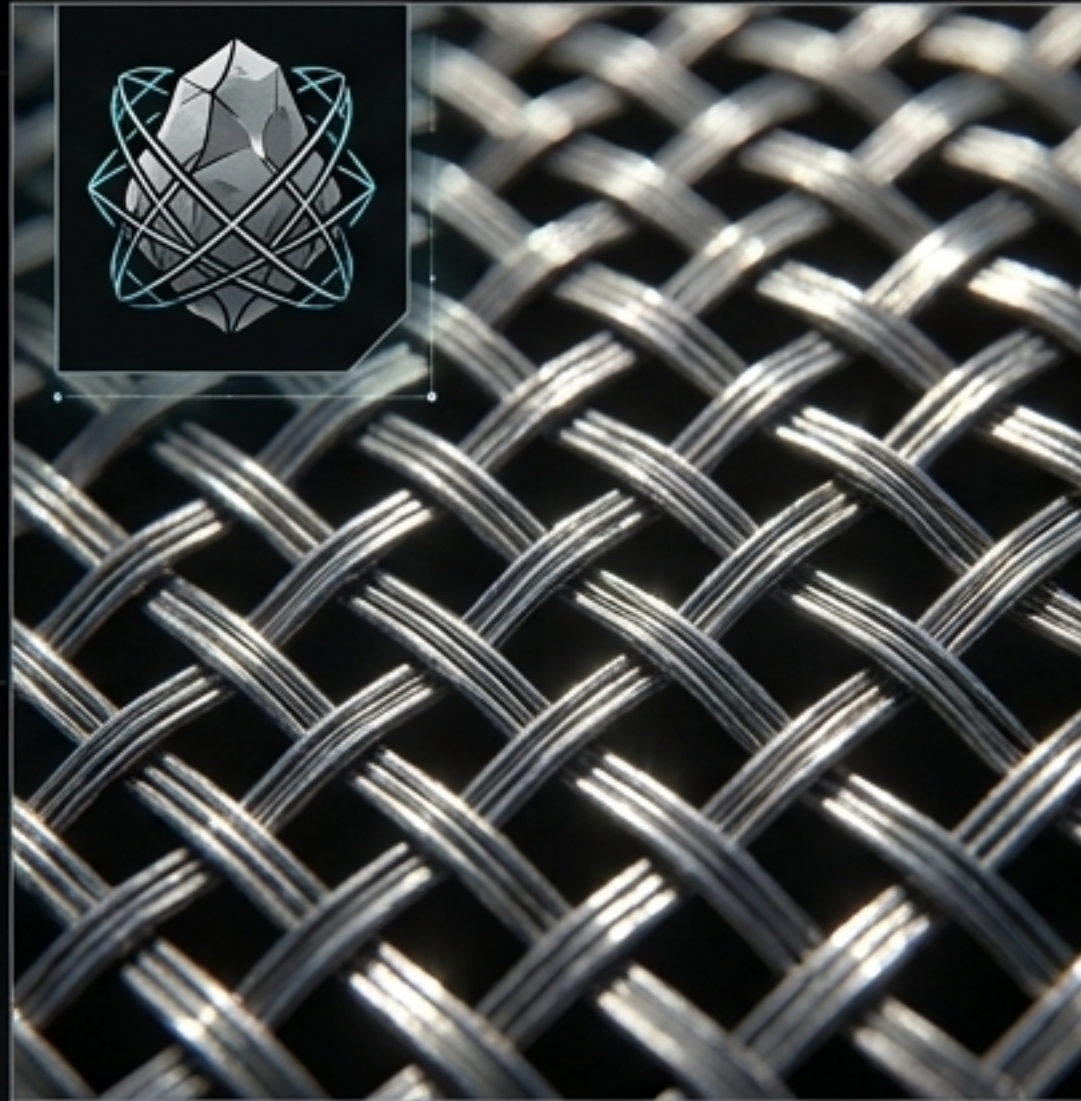
Pre-processing enclosure. Before a single fracture is induced, the body must be encapsulated to prevent high-velocity mass shedding.

The Function:

Converts the open vacuum into a closed, controllable processing vessel.



Structural Containment: The Arsenal Matrix



Column 1: Strap 'n' Bag (The MVP)

High-tension wire/mesh wrapped pre-fracture to control asymmetrical shatter.

Simplicity:
High

Feasibility:
High

IP-ability:
Very High



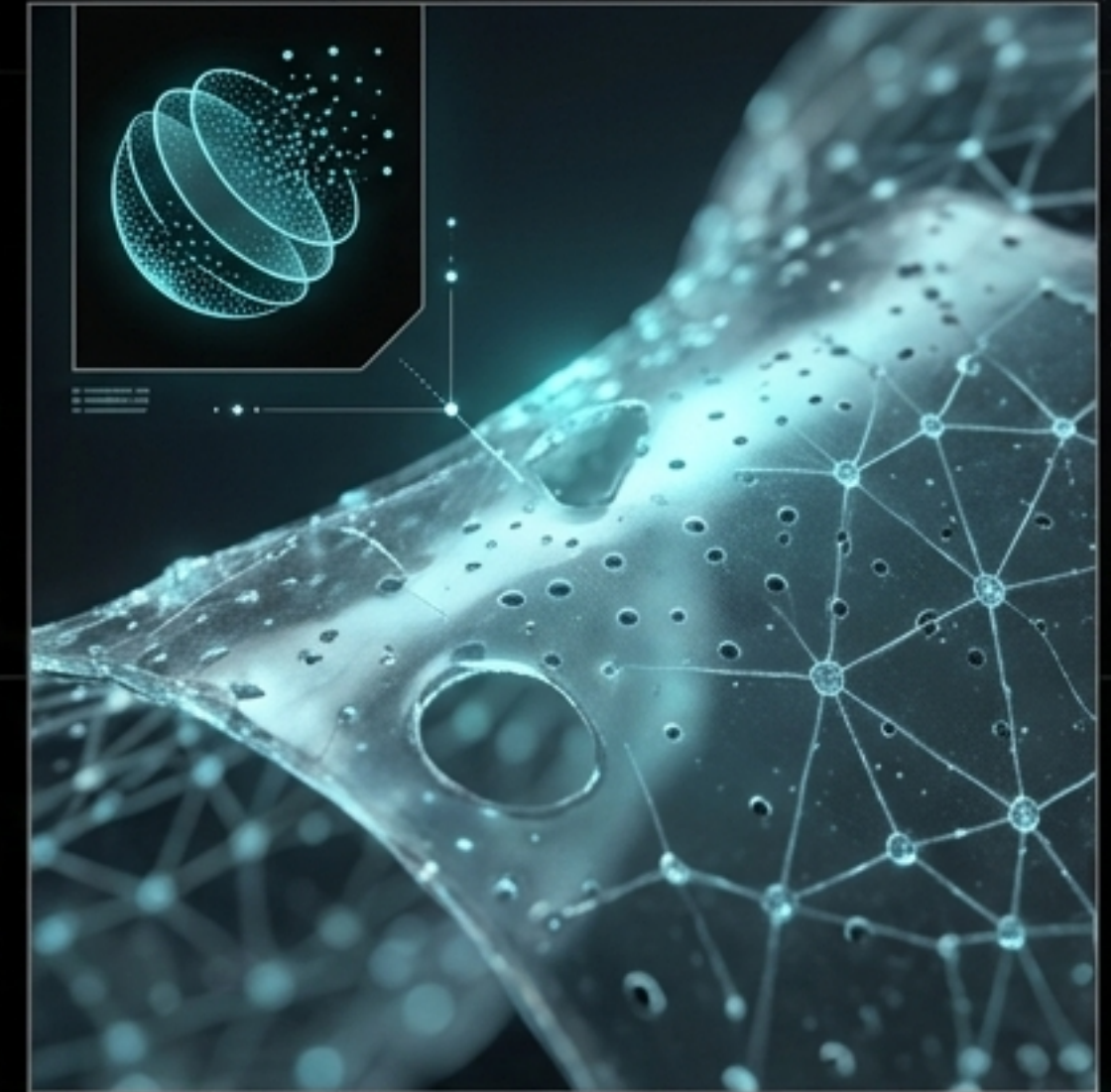
Column 2: Slag Shell Preform

Spin-curing outer regolith into a fused structural shell before deep mining.

Simplicity:
Medium

Feasibility:
High

IP-ability:
High



Column 3: Sacrificial Membranes

Inner tear-mitigation layers that dissolve or disintegrate harmlessly post-processing.

Simplicity:
High

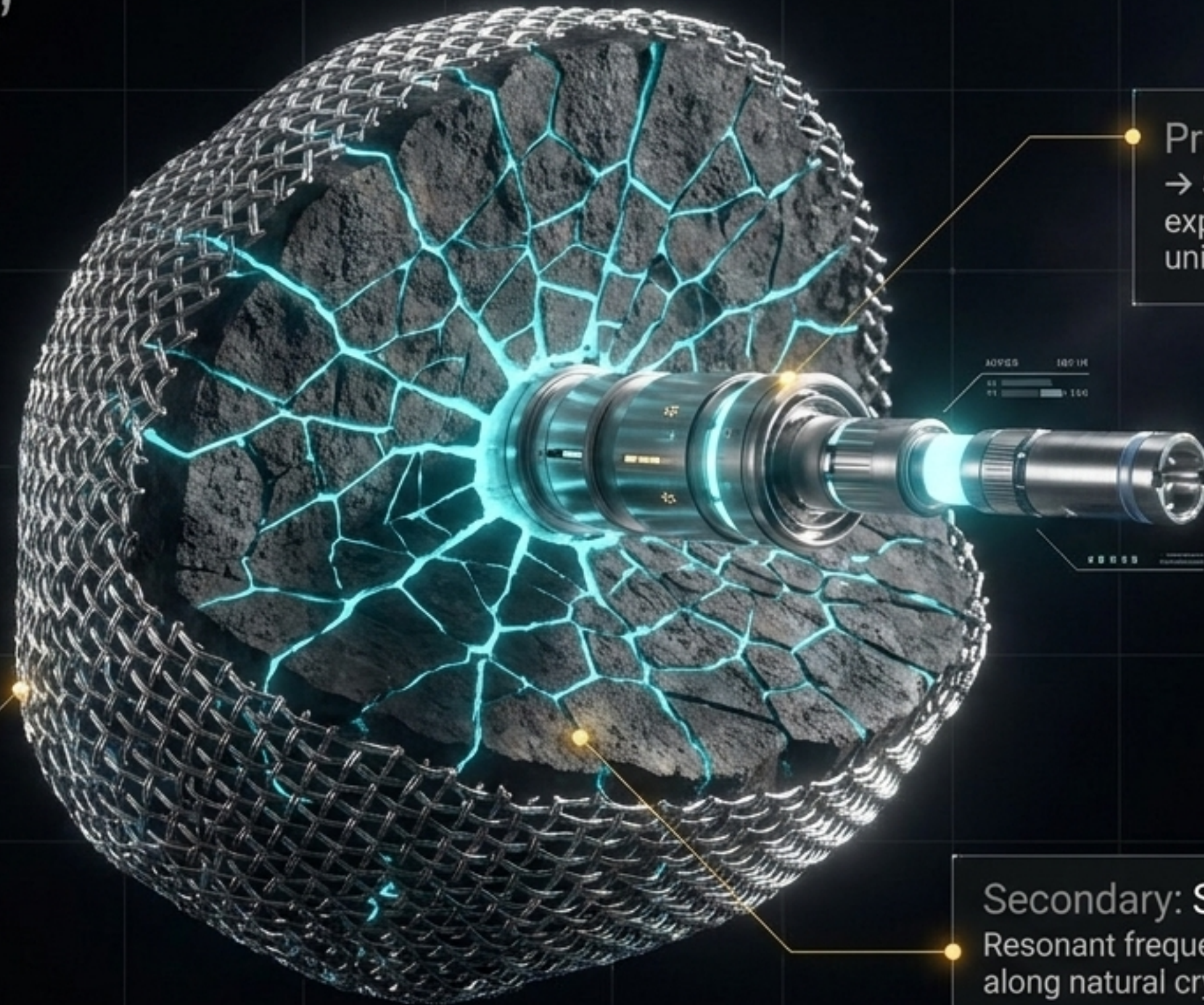
Feasibility:
Medium

IP-ability:
Very High

Disaggregation Mechanics: The 'Wand & Tap'

Philosophy:

How to break a rock gently.
Nixing explosions and thermal
shock for "graceful uncoupling."

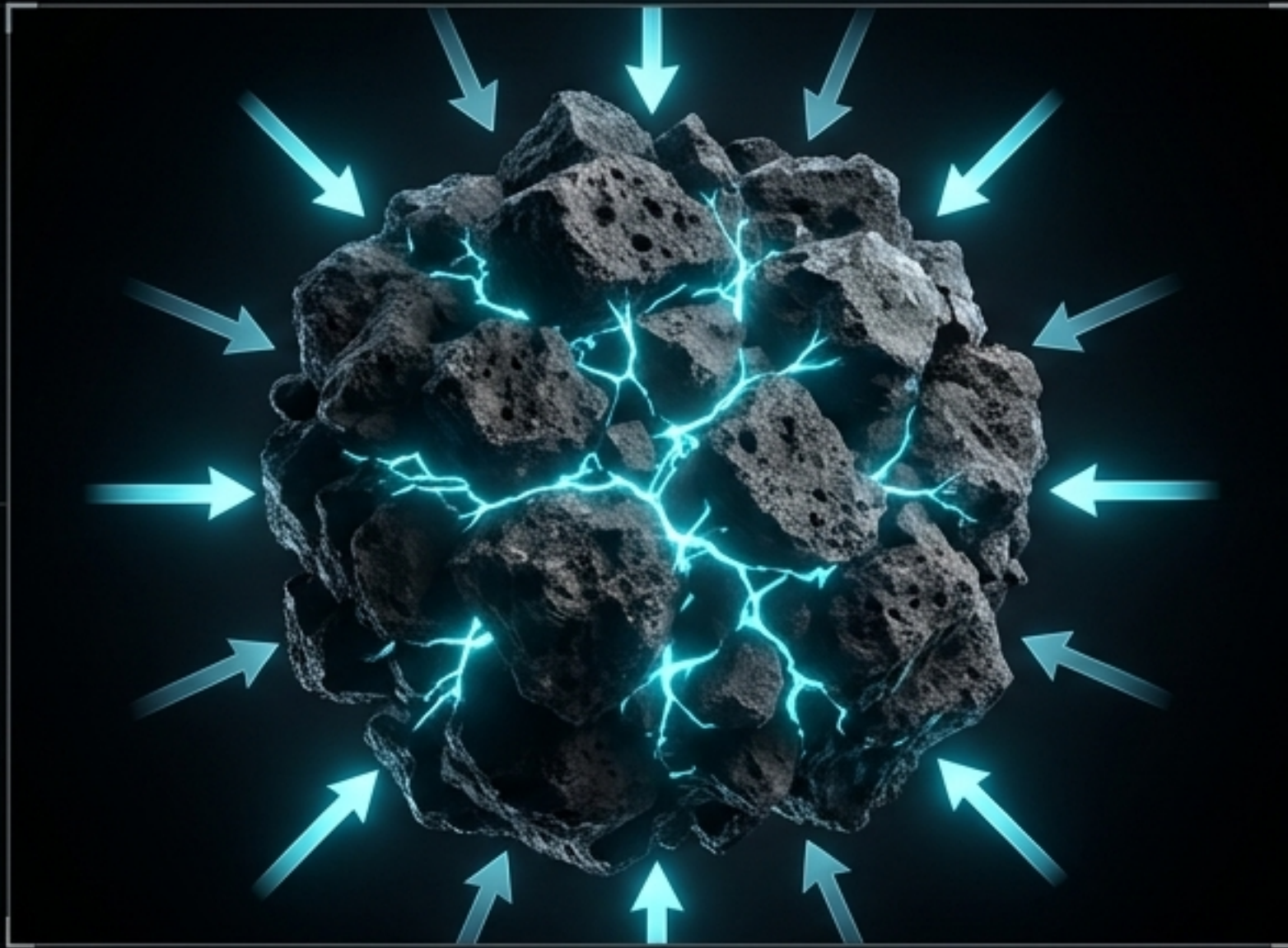


Primary: Drill & Expand
→ Slow-acting chemical/resin
expansion cartridges induce
uniform internal stress fields.

Secondary: Sonic Fracturing →
Resonant frequencies extend fracture planes
along natural crystalline grain boundaries.

The "Snowball & Drumhead" Effect (TCP-Net)

The Snowball (Compression = Cohesion)



The Drumhead (Tension = Transmission)

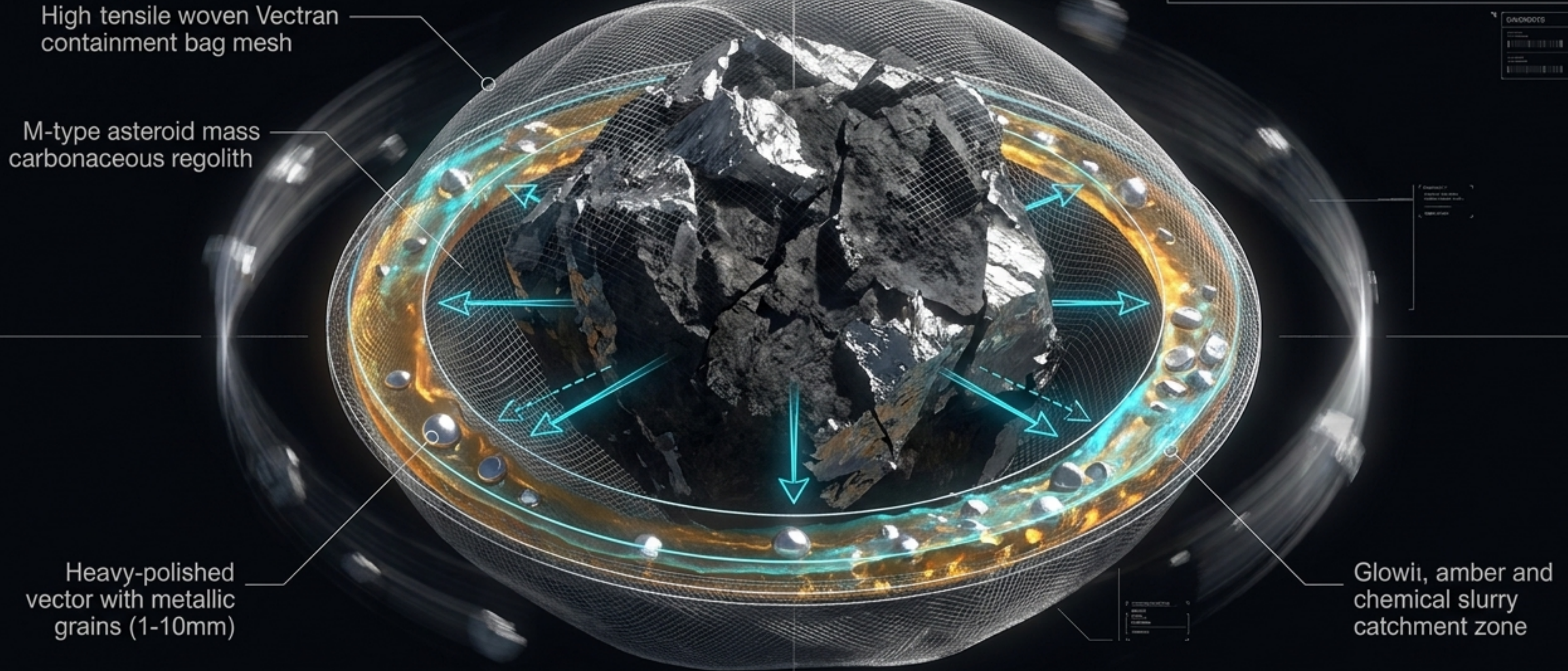


Synthesis: Transient Compaction-Induced Sonic Transmission

The Breakthrough: Compressing a loose pile turns it into a temporary resonant solid. The tensioned net acts as a drumhead, allowing sonic disaggregation to shatter materials that would otherwise absorb the energy.

Continuous Microgravity Fractionation

The 'Cheese Log' Digestion Loop:
Break -> Spin -> Digest.
A pulse-phase reactor system.



The Mechanism: Spin the encapsulated mass at low RPM. Centrifugal acceleration ($a = r\omega^2$) separates 1-10mm particles by mass, pushing high-density PGMs/metals to the perimeter for chemical slurrification and extraction.

The Slag Shell Preform (Circular Architecture)

Waste as Infrastructure:

Turning the asteroid's outer waste layer into a hardened orbital shell.

The Process:

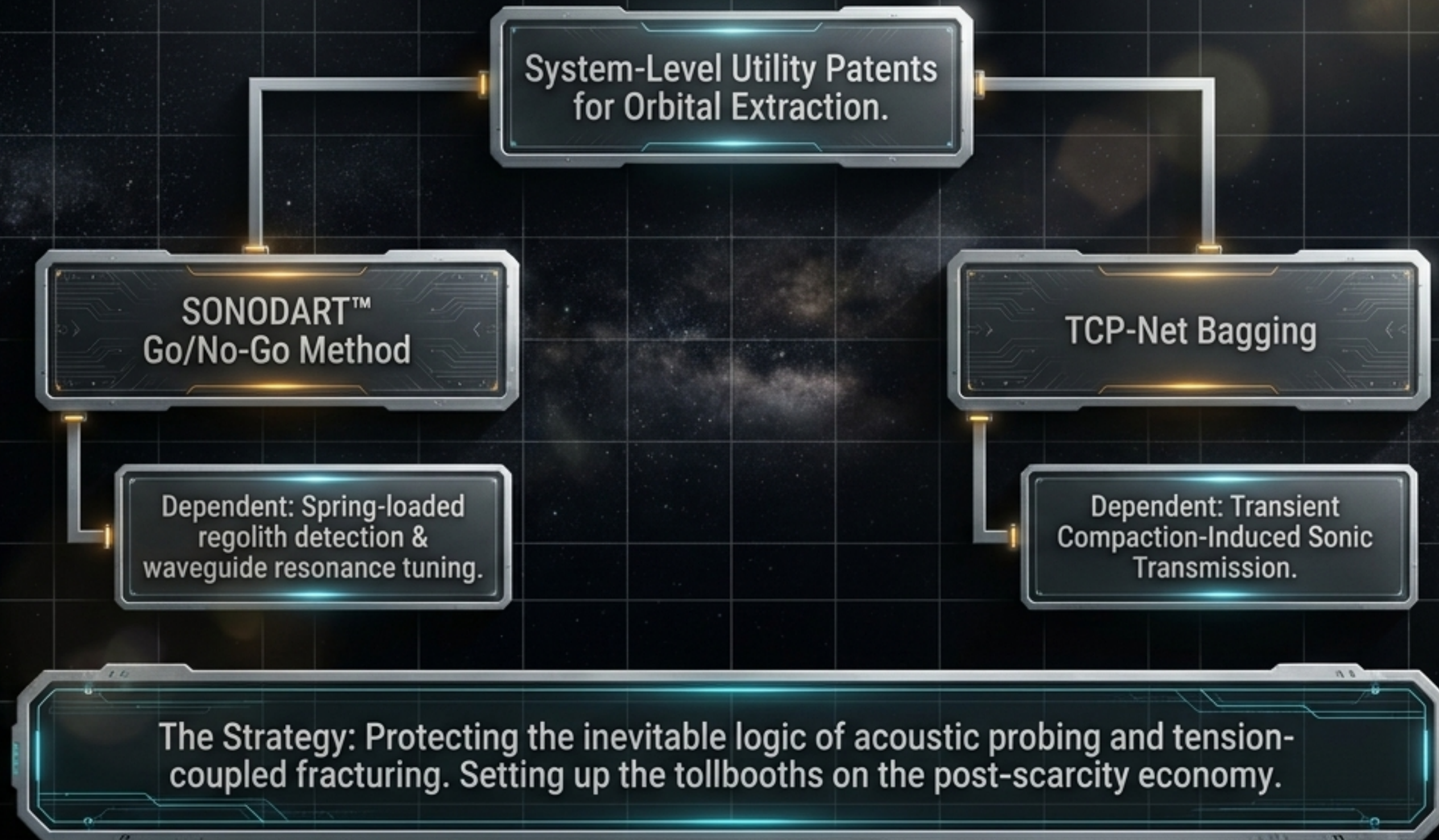
Spin-cured regolith dust + chemical fixatives.

The Payoff:

The byproduct of the "Bag & Break" method becomes the foundation for O'Neill-scale orbital real estate.



IP & Defensive Strategy: The Patent Landscape



The Inevitable Blueprint

The Doctrine: We don't ask
the asteroid if it's mineable.
We strike it. We listen.
We bag it. We break it.

